and eventually PHP is invoked (via Common Gateway Interface, or as an embedded Apache module) with the request data made available via superglobal variables or built-in functions. According to the protocol specification, once the TCP connection is established, an HTTP client should start by sending a request line constructed like so:

- Request method (case sensitive) followed by a single space
- Resource path (commonly known as the URI) and another space
- Protocol version
- ASCII `CR\LF` bytes (carriage-return, line-feed)

These are the very first bytes the web server should receive after the connection is established. Here's an example request line: `GET /index.html HTTP/1.1`, which uses the `GET` method (more on this in a moment) to fetch the resource located via the path `/index.html` (which could be a simple file containing HTML text or something more complex that generates HTML output), using version 1.1 of the HTTP protocol. That's really all there is to the HTTP protocol: roughly human-readable strings of text. I should note here that HTTP/2 onwards uses something called a *binary framing layer*, which encapsulates the request line and headers and turns them into something more machine-optimized for performance reasons, but the heart of the protocol is still the HTTP/1.1 text we're covering here. Note that there is one HTTP method (`CONNECT`) that does not require a resource path.

After the request line come the request header fields, which hold a wide variety of metadata for the request. Headers are used to communicate additional information about the request, such as the client's capabilities and preferences. Some headers are also specific to the response and give detail about encoding, compression, instructions for caching, and so on. A header line is structured in the following way:

- Header name
- Colon character followed by an optional space
- The header's value
- Another optional space

Each header is separated by a `CR\LF` sequence. HTTP 1.1 requires at least the "Host" header (prior to 1.1 there are no request header requirements); here's an example header: `Host: www.example.com`, and the value is, of course, taken from the hostname portion of the URL. As noted earlier, the `Cookie:` header is special: you'll likely have